Project Report

📌 Title

Dynamic Excel Dashboard with VBA Macro Automation and Slicer Control

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📖 Overview

This project demonstrates the creation of an interactive Excel dashboard using Pivot Tables, Pivot Charts, Slicers, and VBA Macros. The main objective was to analyse sales executive performance and automate slicer connections across multiple pivot tables dynamically.

The project simulates a real-world business reporting dashboard where users can filter data region-wise and control which pivot tables are affected by slicers using checkboxes and VBA automation.

🎯 Objectives

- Analyse sales executive performance using Pivot Tables.
- Identify Top 5 and Bottom 5 sales executives.
- Compare target achievement percentages.
- Build interactive charts and dashboards.
- Automate slicer connections using VBA Macros.
- Improve dashboard usability through dynamic filtering.

🛠 Tools & Technologies Used

- **Microsoft Excel**
- **Pivot Tables**
- **Pivot Charts**
- **Slicers**
- **VBA (Visual Basic for Applications)**
- **Macros**
- **Developer Tab Controls**
- **Excel Macro Enabled Workbook (.xlsm)**

📂 Project Workflow

1. Data Analysis using Pivot Tables

Created multiple Pivot Tables to analyse sales performance:

- Top 5 Sales Executives
- Bottom 5 Sales Executives
- Top 5 Executives by Target Achievement %
- Bottom 5 Executives Away from Target %

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2. Dashboard Visualization

Created different chart types for better visualization:

- Pivot Chart for overall sales analysis
- Pie Chart for Top 5 performers
- Line Chart for Bottom 5 performers

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3. Interactive Filtering with Slicers

Added a Region slicer to filter dashboard data interactively.

Initially, the slicer affected only one pivot table. Using the **Report Connections** feature, the slicer was manually connected to multiple pivot tables.

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4. VBA Automation

To avoid manually linking and unlinking slicers repeatedly:

- Added checkboxes using the Developer Tab
- Linked each checkbox to worksheet cells
- Recorded macros for slicer connection actions
- Modified VBA code using IF conditions
- Automated slicer connection/disconnection based on checkbox selection

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⚡ VBA Logic

The VBA macro checks whether a checkbox value is TRUE or FALSE:

- TRUE → Connect slicer to pivot table
- FALSE → Disconnect slicer from pivot table

This makes the dashboard dynamic and user-friendly.

📊 Key Features

- ☑ Interactive Excel Dashboard
- ☑ Automated Slicer Management
- ☑ Dynamic Pivot Table Linking
- ☑ VBA Macro Automation
- ☑ Performance Analysis Dashboard
- ☑ Region-wise Filtering
- ☑ User-Controlled Dashboard Interaction

📈 Business Insights

- Identified top-performing sales executives.
- Detected employees farthest from targets.
- Compared target achievement percentages.
- Improved sales performance monitoring.
- Created an interactive reporting system for decision-making.

🚀 Conclusion

This project demonstrates advanced Excel skills including Pivot Tables, Slicers, Dashboard Design, VBA Automation, and Interactive Reporting. It showcases how Excel can be transformed into a powerful business intelligence and reporting tool using automation techniques.